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Biochar as a sustainable product for remediation of petroleum contaminated soil

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ABSTRACT

Globally, petroleum has been the largest source of energy consumption compared to other energy sources such as natural gas and coal. As a result of high demand for petroleum as an energy source, contamination takes place due to maintenance, production, exploration, storage, transportation and accidental release, yielding in substantial ecological impacts. The contaminations of petroleum in the presence of soil have a negative impact on human health and the development of plant growth. According to this, there is an imperative need to investigate on environmental-friendly remediation strategies to eliminate petroleum contaminants in soil. Among different remediation technologies, bioremediation is recognized as a clean soil treatment technology due to its advantages of safety, no secondary pollution, and economy. For bioremediation of contaminated soil, biochar (BC) applications may provide new solution for polluted soil problems. BC has displayed a good potential to remedy soil contaminants owed to its wide availability of the necessary feedstock, sustainable nature, cost-effective feasibility, high efficiency, large internal surface area and desirable physicochemical surface properties. The usage of BC may include sequestration of carbon, improvement fertility of soil, remediation, as well as recycling of agricultural waste. This paper provides a review on current properties and preparation of BC and its usage as an ameliorant, amendment or sorbent for remediation of contaminated soil. Different research and published papers in this field reveal that the potential of biochar for contaminated soil remediation relies on its physical and chemical properties, which are considerably influenced by the feedstocks, pyrolysis conditions and pyrolysis technologies. The results show that the pyrolysis conditions can affect the BC structure and composition and thus its adsorption capacity. Pyrolysis temperature influences the adsorption characteristics of BC by changing the surface functionality, surface area, pore distribution and mineral concentration of BC. It is important to produce BC at optimum pyrolysis temperatures to maximize inorganic and organic contaminant adsorption. The results also indicate that BC obtained through microwave pyrolysis has better yield and quality than those produced from conventional pyrolysis.

1. Introduction

Nowadays, rapid growth in urbanization, population and living standards play a crucial role in increase of biomass waste generation (e.g., agricultural/forestry and food wastes) calling for a sustainable solution [1,2]. Currently, different environmentally acceptable strategies have been discovered for the sustainable waste management dealing with this problem [3,4]. Due to the importance of environmental and economic indicators, the conversion of biomass into BC is a promising solution for waste management [5,6]. Several kinds of biomass can serve as source materials of BC, for example, animal manure, woodchips, crop residues [7,8]. In this sense, enhancing recycle of agricultural and forestry wastes is also a noticeable advantage of BC [9,10]. These diverse potential benefits, incorporated with the fact that BC produced from a range of biomass can potentially be a relatively cost-effective and environmental beneficial tool for environmental remediation, stimulates increasing research on BC [11,12]. A relationship among BC, waste recycling and soil remediation is shown in Fig. 1.

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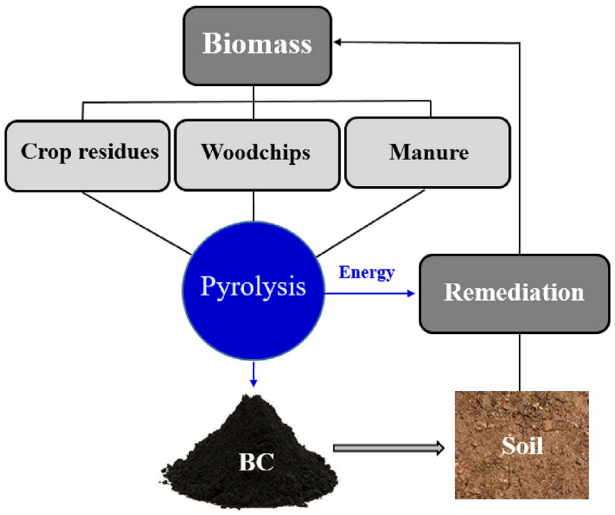


Fig. 1. BC, waste recycling and soil remediation.

BC is the porous, carbonaceous material, obtained from conversion of organic materials via thermochemical (e.g., pyrolysis, gasification) and hydrothermal processes [13,14]. BC is increasingly receiving attention and highly recommended as soil amendment because it cannot only mitigate climate change by sequestrating carbon from atmosphere into soil but also improve soil properties and enhance soil fertility by improving microbial activity and moisture and nutrients retention hence increasing crop productivity [15,16].

Sorption and interactions between active functional groups on the BC are the main mechanisms to remove organic and inorganic pollutants from soil, respectively because of existing porosity of BC it can hold water in pores and increase soil water holding capacity, therefore lead to improvement of soil fertility [17,18]. In contrast with other porous carbon materials [19–21], porous BC usually possesses abundant surface functional groups, as Fig. 2 depicts. These surface functional groups and porosity of BC are very helpful for adsorbing inorganic and organic contaminants.

The existence of micro-, meso- and the macropores in solid adsorbent materials exhibiting porosity is of great attention in real applications because of the characteristics of mass transfer. According to the International Union of Pure and Applied Chemistry (IUPAC) recommendations, the micropores are defined as pores with diameter not exceeding 2 nm, mesopores are pores with diameter between 2 and 50 nm, and macropores represent pores with diameter greater than 50 nm (Fig. 3) [23–25].

Modifying microbial life and nutrient in the soil are the other possible mechanisms for enhancing soil fertility [26,27]. Pyrolysis, a process in which biogenic waste is heated at a temperature range of 250–850 °C in the absence of oxygen or in the presence of a limited amount of oxygen, is one of the most promising technologies for BC production [28,29]. Syngas and hydrocarbon-rich bio-oil are the other products of pyrolysis [30]. Pyrolysis can be divided into three principal categories: slow pyrolysis, fast pyrolysis, and flash pyrolysis. Biomass particle size, retention time, reaction temperature, heating rate and time are the main factors affecting the classification of pyrolysis and its product distribution [31–34]. The operating conditions and products yields for pyrolysis processes are collected in Table 1.

2. Factors affecting BC quality

The Feedstock properties and production conditions are the two main classifications that determine the properties and yield of BC. Břendová et al. [36] investigated the pyrolysis condition on specific surface area and yield of BC. They reported that properties of BC were affected by both plant material and pyrolysis temperature. The results showed that, the final temperature raised, the surface area improved but the yield of BC reduced. The largest surface area was found at BC from wood mixture and the lowest one from hay of grass.

Igalavithana et al. [37] reported the BC properties produced from different processes and the effect of different conditions on its surface area (Fig. 4). They reported that the BC production temperature could significantly change its composition.

The modifications of BCs or their feedstocks can alter their physicochemical properties, pyrolysis conditions can also affect the BC structure and composition and thus its adsorption capacity. Pyrolysis temperature influences the adsorption characteristics of BC by changing the surface functionality, surface area, pore distribution and mineral concentration of BC. It is important to produce BC at optimum pyrolysis temperatures to maximize inorganic and organic contaminant adsorption. Jimenez et al. [38] studied the effect of pyrolysis temperature on the properties and adsorption capacity of BC. They reported that high temperature BC has a higher surface area, pores previously blocked by unpyrolyzed organic matter are unblocked, and there are more aromatic groups and less oxygen containing functional groups on the BC surface

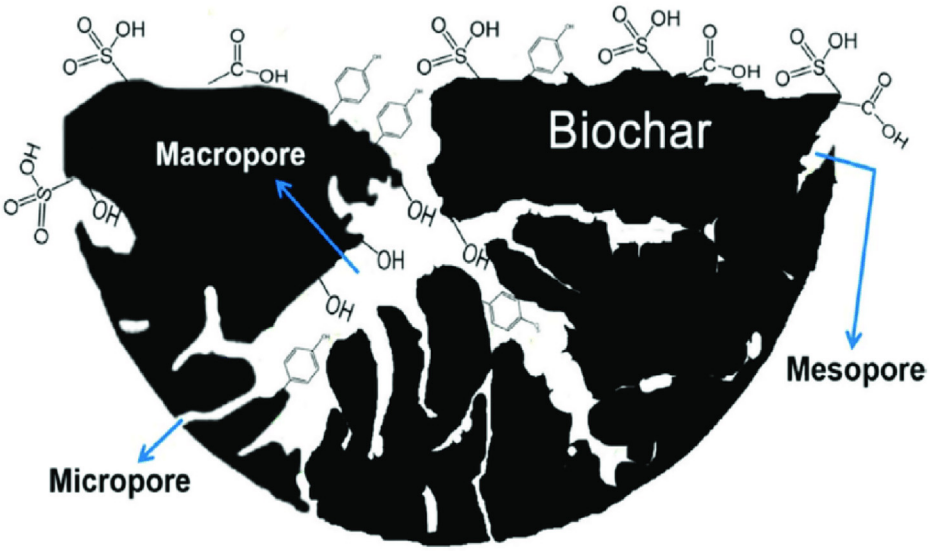


Fig. 2. Schematic of porous BC containing various functional groups [22].

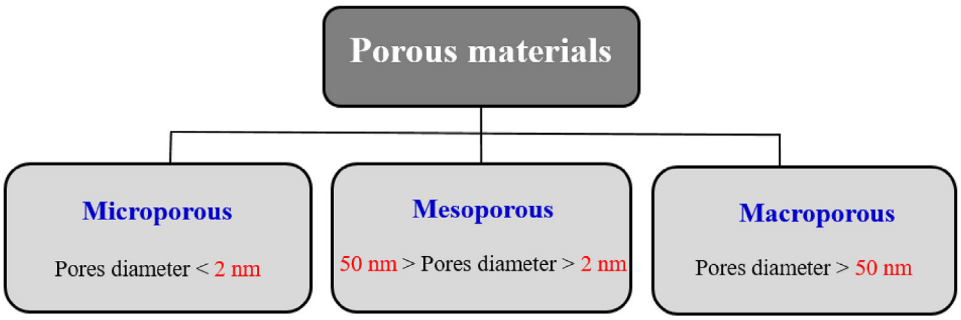


Fig. 3. Classification of porous material according to pore size.

Table 1
The operating conditions and products yields for pyrolysis processes [31,35].

Group	Residence time	Temperature (°C)	Heating rate (°C/S)	Particle size (mm)	Char (%)	Liquid (%)	Gas (%)
Slow pyrolysis	>30 min	400	0.1–1	5–50	35	30	35
Fast pyrolysis	10–20 s	500	1–200	<1	20	50	30
Flash pyrolysis	1 s	500	>1000	<0.2	12	75	13

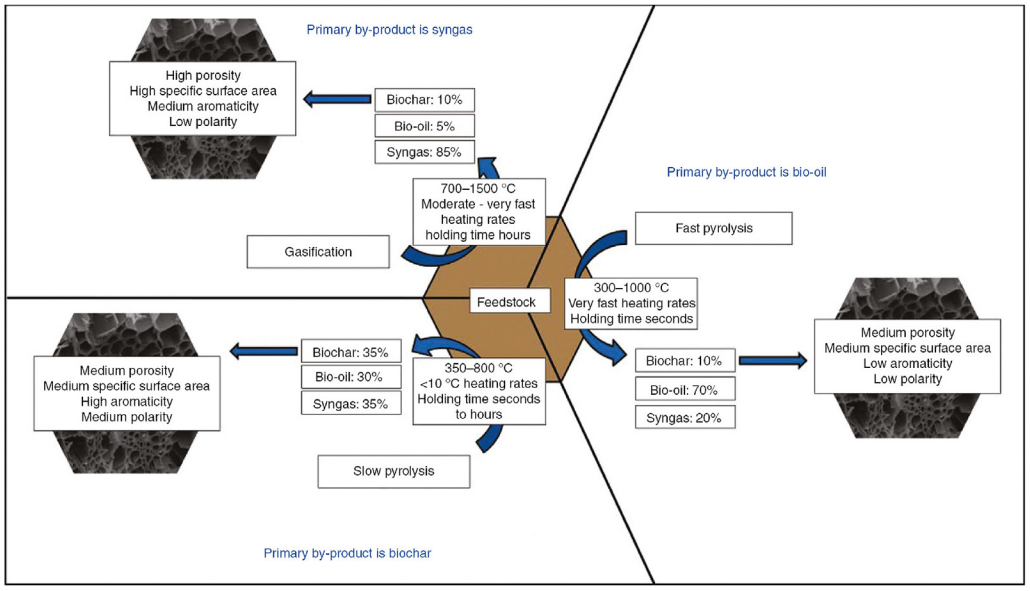


Fig. 4. The properties and differences of the BCs produced from different processes [37].

(Fig. 5).

Zhao et al. [39] studied the influence of a wide variety of diverse feedstock and a vast range of production temperatures on the properties and yield of BC. From the above data it becomes evident that the yield and properties of BC were influenced by both feedstock properties and production conditions. In addition to this, there are various factors which affect quality of BC derived from different biomass. Liao et al. [40] by analysis of 16 quality indexes of 24 BCs found that BC parameters such as cation exchange capacity (CEC), pH, specific surface area, hydrogen atom content, phosphorus and potassium content and maximum adsorption capacity of NH4+-N are most important in evaluating the quality of BC.

2.1. Pyrolysis conditions

Biomass feedstock, time and temperature during pyrolysis are the most vital factors that affecting physicochemical properties of BC such as surface complexation, size of the pore, CEC, water holding capacity, ash content, volume and distribution [41]. Ash content of BC is more dependent on feedstock composition than pyrolysis temperature, while volatile solids (VS), surface area and carbon fixed content of BC are mainly affected by pyrolysis temperature [36,42]. For example, the effect of pyrolysis temperature on the carbon content of corn Stover BC was evaluated by Rafiq et al. [43]. Their experimental result revealed that with increasing pyrolysis temperatures carbon content of BC increased from 45.5% to 64.5%. Titiladunayo et al. [44] investigated the effects of temperatures (400, 500, 600, 700, and 800 °C) during pyrolysis on BC yield produced from biomass (Elaeis guineensis (palm kernel shell), Milicia excels (iroko wood) and Afzelium Africana (APA wood)). They reported that heating rates and temperature are considered as main parameters affecting BC production. As the temperature severity across species enhances, the amount of charcoal obtained by the process reduces. The industrial application qualities of the obtained charcoal regarding lower moisture content, volatile matter, ash and sulphur contents, high fixed carbon, content and percentage carbon, were achieved at high heating rate and temperature for all species. Zhao et al. [45] reported that BC quality have improved with increasing temperature.

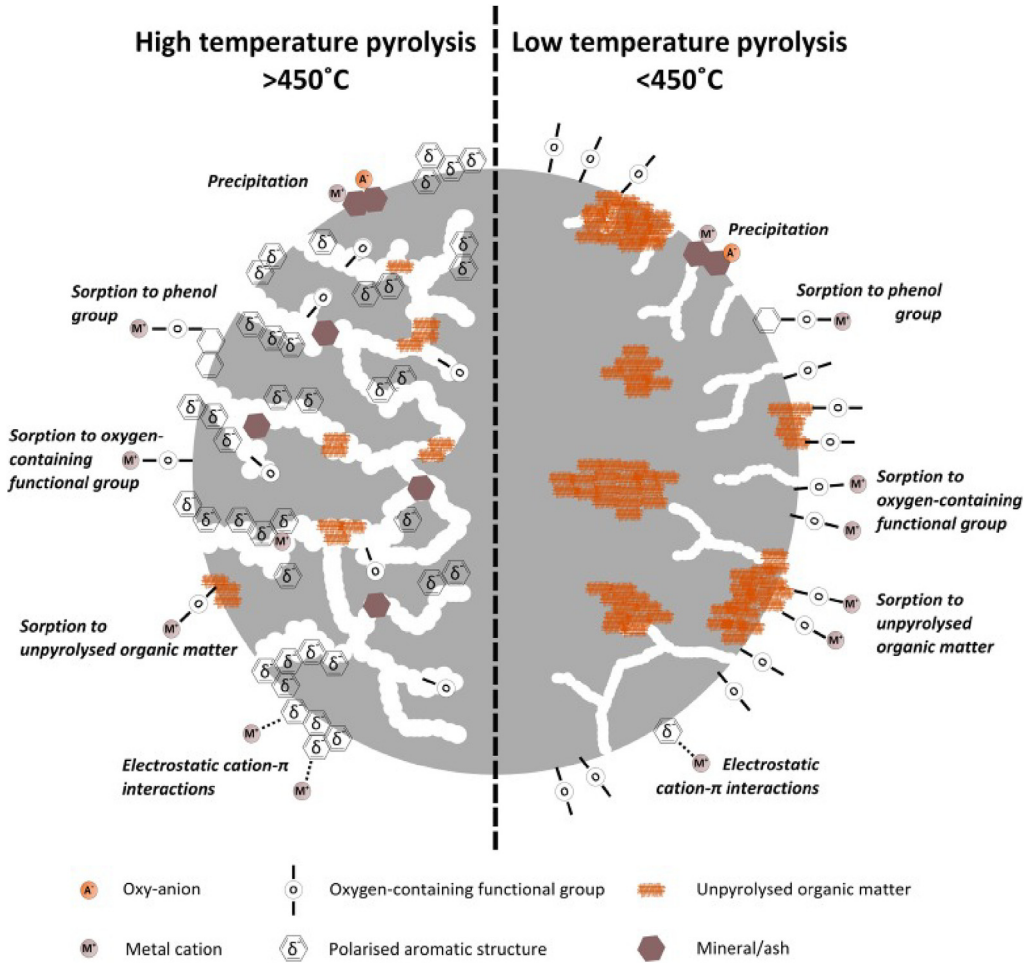


Fig. 5. Conceptual model identifying the mechanisms of metal cation and oxyanion. Adsorption to BC pyrolyzed at low temperature (<450 °C) and high temperature (>450 °C) [38].

They used industrial waste lignin as feedstock for slow pyrolysis to study the influences of temperature (300–700 °C) on yields and character of pyrolytic products (bio-oil, bio-char and gas). The results proved that pyrolysis temperature played an important role on the conversion of waste lignin into high value-added products. BC was the dominant product of waste lignin, whose quality improved with increasing temperature, while the best combustion reactivity was focus at 500 °C. The optimum condition for bio-oil formation was at 600 °C, where contained a mass of phenols. Gas yield significantly increased effected by temperature, especially CH_4 [45]. In continue to related investigation, the effects of pyrolysis temperature (400, 600 and 800 °C) on adsorption capacity for aromatic contaminants (guaiacol, anisole and phenol i.e.) of magnetic BC formed by thermal pyrolysis of $Fe(NO_3)_3$ and distillation residue achieved from rice straw pyrolysis oil were estimated by Li et al. [46]. It is found that with increasing pyrolysis temperature the degree of carbonization of magnetic BCs increased. Owing to the blockage of channels in BC during pyrolysis at 800 °C, the largest textural properties like pore volume and surface area of the magnetic BC occurred at the pyrolysis temperature of 600 °C. Adsorption of aromatics were carried out using batch adsorption approach. The adsorption experiments were performed over the initial aromatics concentration range from 3 to 35 mg/L and about 0.0090 g magnetic BC was used as adsorbent. According to the removal results, the used adsorbent could be magnetically separated and the adsorption capacity of anisole on magnetic adsorbents was better than that of guaiacol and phenol. The properties of magnetic BC, such as iron oxide particles, graphene-like-structure, pore volume, surface area and aromaticity, displayed outstanding impacts on the uptake potential of aromatic pollutants. Wang et al. [47] studied converting industrial waste cork to BC as Cu (II) adsorbent through slow pyrolysis. They assessed that the cork-based BC produced under conditions of higher pyrolysis time and temperature, possess larger specific surface area, lower polarity, higher aromaticity and enhanced Cu ion uptake ability. In addition to this, after thermal decomposition, a large number of pores are created in the cell wall and these cells are attached to each other via open pores. To investigate the effect of pyrolysis temperatures on physical and chemical characteristics of BC for soil amendment, rice straw was pyrolyzed at temperatures varying between 400 and 600 °C by Yakout, S. M [48]. In this study, organic removal was determined by shaking 0.1 g of BC to 50 mL aqueous solution of 100 ppm concentration in a flasks for 24 h. They reported that high pyrolysis temperatures cause condensation reactions, that results in reducing of BC polarity and raising of aromaticity. Hydrophilicity and alkalinity-acidity of BC is recognized as being one of the most key characteristics that impress the adsorption behavior of the solution. High pyrolysis temperatures produce BC with higher carbon sequestration potential characteristics when used as soil amendment. Ash content, acidity, yield, acidity, oxygen and total hydrogen contents diminished when pyrolysis temperature increased. As pyrolytic volatilization via pyrolysis total carbon, nitrogen and basicity decreased. The pyrolysis temperature at 500 °C indicates that the textural properties of BC derived from rice straw such as micro-mesopore volume and surface area are at their maximum [48].

Zhao et al. [49] investigated the influence of pyrolysis temperature, residence time and heating rate on rapeseed stem derived BC. They found that pyrolysis temperature was the most influential pyrolysis parameter

and in comparison to other obtained BCs, the rapeseed stem BC commonly had further pH values and lower O/C ratios, ash content as well as yield. Sarkar et al. [50] appraised the different pyrolysis process conditions of South Asian waste coconut shell. At their research, the gas production raised from 15.4 to 18.3 wt% as the temperature raised from 400 to 800 °C. In the other study, biomass feedstocks of mixed larch, spruce chips and pine, softwood pellets were used to produce BC at a certain temperature ranged from 350 °C to 550 °C to find out the relation of BC production conditions with the yield and its long-term stability. The experimental data of this study demonstrate that the performance of stable BC fraction is not related to the temperature, in spite of the improve in the stability of BC with raising pyrolysis temperature [51]. Also, the effects of different heating rates and pyrolysis temperatures on the properties and yield of BC achieved from pyrolysis of solid olive mill waste (pomace) were investigated by Hmid et al. [52]. The C content and BC aromaticity increased with increasing the heating rate and pyrolysis temperature while the surface functional groups and the BC yield were diminished. The highest BC performance was obtained by low heating rate and low pyrolysis temperature. The results of the other experiment conducted by Mimmo et al. [53], confirmed that physical and chemical characteristics of miscanthus (Miscanthus × giganteus) BC and its short-term stability are affected by pyrolysis temperature. Angın, D [54]. studied the impacts of heating rate and pyrolysis temperature on the physicochemical properties and yield of BCs obtained from safflower seed press cake and reported that the BC yield and quality depend on the applied temperature. The results showed that BC including higher fixed carbon content (80.70%), carbon percentage (73.75%), and higher heating value (30.27 MJ kg^{-1}) compared to the original feedstock (SPC) and the low content of volatile matter (9.80%) produced when pyrolysis temperature was 600 °C. Manya et al. [55] explored the effect of peak temperature, pressure, and pyrolysis atmosphere during the slow pyrolysis of three biomass sources (vine shoots, corn stover and two-phase olive mill waste). Their results demonstrated that the significant increase in the yield of produced gas was observed when pressure was raised. Also by using CO_2 as carrier gas, two valuable products can be obtained. These products are a BC with an appropriate potential to sequester carbon, and a produced gas with an appropriate composition to the purpose of energy recovery.

Biomass pyrolysis is a thermal degradation process in the absence of air or oxygen to produce BCs [56]. A novel, emerging technique is to use microwaves (MW) as the heating mechanism [57]. MW technology is a type of electromagnetic radiation with wavelengths that vary from as short as 1 mm to as long as 1 m. The MW electromagnetic spectrum ranges between infrared and radio frequency, resulting to a frequency range from 300 MHz to 30 GHz [58]. Generally, the difference between MW and conventional (CV) heating can be mainly assigned to their different heating mechanisms (dielectric heating vs. convective and conductive heating) and to the occurrence of a hot spot (spark) generated by MW heating [59]. In MW pyrolysis, the temperature at the center of the feedstock is higher than the materials surface and surrounding as illustrated in Fig. 6.

The schematics of the pyrolysis reactors with CV heating and MW heating are presented in Fig. 7.

Several researchers focused on MW pyrolysis over the CV pyrolysis process [62,63]. The advantages and disadvantages of MW pyrolysis and CV pyrolysis are summarized Table 2 [64,65].

MW pyrolysis has a lot of advantages such as wet biomass can be used directly without drying, no size reduction, higher quality of products, and process contingency [66]. All these will not only improve the energy efficiency for the process, but also reduce the capital cost, such as feed dryer, size reduction, and feed storage. Another key advantage of the MW heating process over CV heating method is the nature of fast internal heating by MW irradiation [67]. MW energy deposition in the dielectric loss mode of heating can cause spatially uniform heating. Biomass fast pyrolysis can be achieved by MW radiation. Miura et al. [68] investigated the difference in pore characteristic and carbon-like adhesive material in

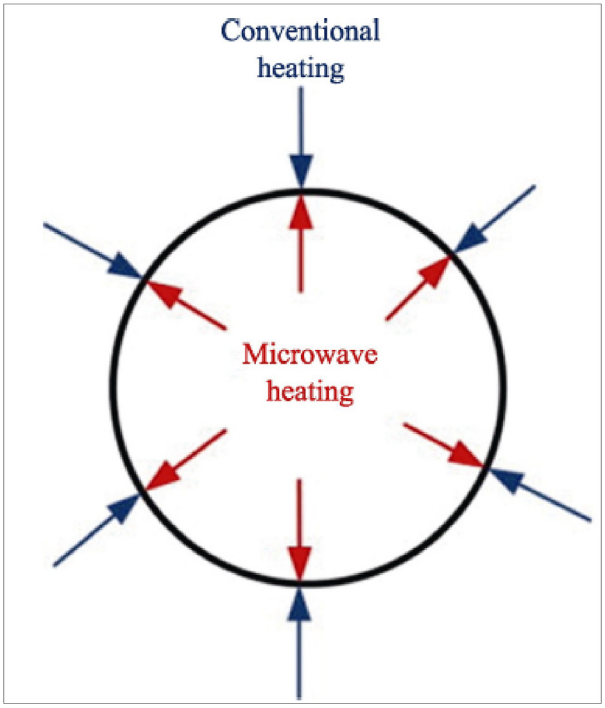


Fig. 6. MW and CV heating nature [60].

the BC matrix during CV and MW pyrolysis. It was observed that the carbon particle was deposited in the micropores of BC during CV pyrolysis; this is because the rate at which the volatiles escaped from the biomass was slow during pyrolysis. In contrast, clean and more open porous structures were found in MW pyrolysis, primarily because of the high escaping rate of volatile matter.

Comparatively, lower temperatures as well as a more homogeneous and selective heating generate a higher quality BC. Budarin et al. [69] examined the MW decomposition of cellulose. In comparison with CV pyrolysis, MW of cellulose can be obtained at much lower temperature (180–200 °C) and the amorphous region showed thermal degradation as low as 180 °C producing a useful BC product.

Furthermore, MW pyrolysis reduces hazardous product formation and minimizes emission pollutants, making the technique environmentally friendly. BC obtained through MW pyrolysis has better yield and quality than those produced from CV pyrolysis [70].

Generally, different parameters of MW pyrolysis will lead to varying yields and qualities of produced BCs. To achieve the maximum conversion yield and highest quality, parameters of MW pyrolysis should be optimized. The most important parameters in MW pyrolysis reaction processes include: (i) type and size of input biomass, (ii) MW output power, (iii) water content and moisture of input biomass, (iv) temperature and time of reaction, (v) MW type (vi) type, size, and amount of MW receptor, (vii) mixing intensity, (viii) type and design of reactor, and (ix) Type and flow rate of carrier gas [66,71].

2.2. Biomass feedstocks

In the recent years, the effect of feedstock differences on BC characteristics has been widely studied. Different sources of agricultural crops or agricultural by-products have been employed as biomass feedstocks for BC production by Yang et al. [42] and Xie et al. [72]. These sources have composed of three main components lignin, cellulose and hemicellulose (Table 3).

Cellulose and hemicellulose are more volatile and almost all of them pyrolysis at 350 °C, while the lignin is the most difficult to volatilize and pyrolysis at 500 °C [74]. The highest bio-char yield produces by pyrolysis

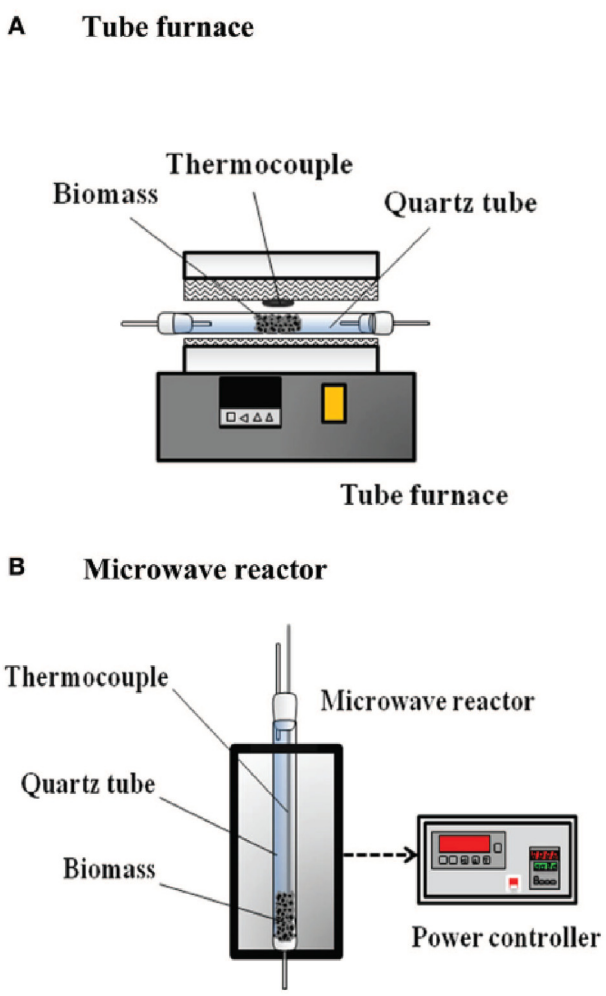


Fig. 7. Schematics of pyrolysis reactors with (A) CV heating and (B) MW heating [61].

Table 2
The advantages and disadvantages of MW pyrolysis and CV pyrolysis.

	CV pyrolysis		MW pyrolysis	
	Advantage	Disadvantage	Advantage	Disadvantage
1	Flexibility of feedstocks and products	Lower quality products (e.g. formation of PAHs)	No size reduction and drying for biomass; Flexibility of feedstocks and products	Temperature measurement
2	Well developed and easy scale-up	Energy consumption due to higher temperature (>600 °C)	Energy savings at significantly reduced processing temperatures (150–300 °C)	Economics of the process scaling-up
3	Possibility of continuous processing at large scale	Lower quality products (e.g. formation of PAHs)	Possibility of continuous processing with multiple units	Large scale inhomogeneneities/reproducibility issues in products

at moderate temperature (500 °C) from biomass feedstocks with high lignin content. To determine characterization of the pyrolysis products with an emphasis on the bio-char and bio-oil, the wastes of palm shell, mesocarp fiber, and empty fruit bunches were pyrolyzed under certain conditions of contact time (60 min), nitrogen flow rate (2 L/min) and

Table 3
Hemicellulose, cellulose and lignin content of biomass feedstocks [35,73].

Feedstock	Lignin (%)	Cellulose (%)	Hemicellulose (%)
Wood	25–30	35–50	20–30
Hardwood barks	30–55	22–40	20–38
Softwood barks	30–60	18–38	15–33
Sugarcane bagasse	23–32	19–24	32–48
Nut Shell	30–40	25–30	25–30
Rice straw	18	32.1	24
Wheat straw	15–20	33–40	20–25
Barley strew	14–19	31–45	27–38
Corn stover	7–21	38–40	28
Algae (green)	–	20–40	20–50
Sorghum straw	13	32	24
Sweet sorghum Bagasse	14–22	34–45	18–24
Cornstalk	3–5	39–47	26–31
Grasses	10–30	25–40	25–50
Chemical pulp	2–10	60–80	20–30
Leaves	–	15–20	80–85
Walnut shell	52.3	25.6	22.7
Hazelnut shell	42.9	28.8	30.4

temperature (500 °C). Experimental data show that the among the biomasses reported palm shell contained the highest level of lignin revealed the highest levels of bio-char yield and fixed and elemental carbon in the raw and bio-char form while palm oil wastes yielded more bio-oil than bio-char or non-condensable gases [75]. In addition, pyrolysis product yields have been affected by the ratios of volatile matter, fixed carbon, ash content (inorganic materials) and moisture of biomass feedstocks. Biomass feedstocks with high volatile solids components produce more bio-oil and syngas during pyrolysis whereas fixed carbon increases the BC production [35]. Also because of more energy need for pyrolysis the particle sizes and moisture contents of feedstock are important too [76]. Friedl et al. [77] reported that feedstocks composed of low mineral and nitrogen contents (such as wood and biomass from low-cost agricultural by-products) are suitable for bio-oil and syngas production. Therefore, both of physical and chemical characteristics of biomass feedstocks have influence in the pyrolysis product yields [35]. A comparison of physicochemical properties of biomass feedstocks for BC production is tabulated in Table 4.

Rajapakasha et al. [81] characterized ten BC samples produced from different feedstocks. They illustrated that Protein-/tannin-like component in BC was mainly influenced by feedstock.

3. Remediation of petroleum contaminated soil

The presence of petroleum hydrocarbon (PHC) contaminants in soil causes significant environmental impacts and poses a substantial hazard to both human and other forms of live in the polluted environments [82]. PHC contaminants characterize vast majority of compounds and by-products that are classified as priority environmental pollutants such as saturated hydrocarbon (SHC), aromatic hydrocarbon (AHC) and non-hydrocarbon (NHC) compounds [83,84]. The SHC molecular structure consists of and carbon hydrogen bond and carbon-carbon bond, which is easy to be degraded. In addition, its boiling point is relatively low, so the SHC can gradually disappear from the soil in the way of volatilization and photosynthesis. However, the molecular structure of AHC is quite complex. The complex benzene ring and its higher boiling point greatly increase the difficulties of being removed from the soil [85,86]. The polycyclic aromatic hydrocarbon (PAH) is the typical persistent organic pollutants (POPs) that widely exists in various environmental systems, and the 16 kinds of PAHs have been included in the list of priority control pollutants by Economic Partnership Agreements (EPAs) of the United States and European Community [87,88]. Among the three kinds of hydrocarbon compounds, NHC compounds possess the maximum carbon number. They are insoluble in water, and the fusion and boiling point are quite high, so they are most difficult to be removed from the soil, resulting in obvious environmental toxicity and

Table 4
Physical and chemical properties of various biomass feedstocks [35,78–80].

	Feedstocks					
	Wood	Sugarcane	Wheat straw	Rice straw	Barley strew	Maize strew
Physical Properties						
Density (kg/m^3)	1186	141.2	1233	200	210	–
Moisture Content (%)	20	10.4	16	6	30	6.24
Ash Content (%)	0.4–1	16.4	4	4.3	6	8.90
Volatile Matter (%)	82	74	59	79	46	76.19
Fixed Carbon (%)	17	13	21	10.7	18	8.67
Chemical Characteristics						
Carbon (%)	51.6	43.2	48.5	48.31	45.7	49.6
Hydrogen (%)	6.3	6.7	5.5	5.32	6.1	6.02
Oxygen (%)	41.5	33.2	3.9	34.63	38.3	42.9
Nitrogen (%)	0.1	0.3	0.3	1.23	0.4	0.68

mutagenicity. Therefore, the natural soil residual poison composition is mainly PAHs and NHC compounds [89]. It should be pointed out that NHC composition contains a small amount of heavy metal elements and some studies have reported that the drilling fluid also contains heavy metal pollutants, with high toxicity, persistence, and refractory [90]. For all the above, used and abandoned sites contain soils contaminated by PHCs. These would have resulted from spills and leaks of gasoline, fuels, kerosene, diesels, lubricating oils and similar products during the years of industrial activities at the site.

Soil pollution proliferations caused by PHCs which have serious threats to human beings, animals, and plants in the environment [91]. Therefore, the remediation of petroleum-contaminated soil is a very critical and essential topic and has attracted extensive and increasing research interest [92]. Nowadays, many technologies are available to deal with petroleum-contaminated soils, such as soil washing, solvent extraction, incineration and in-situ thermal desorption, chemical oxidation, electrokinetic remediation, vermiremediation, bioremediation, and integrated approaches [92–94]. However, most of these techniques lack public acceptability due to their high expense and secondary pollution. Attentions have been paid to alternative bioremediation technology to enhance the degradation of PHCs in soil [95].

Bioremediation is recognized as a clean soil treatment technology due to its advantages of safety, no secondary pollution, and economy [96]. The treatment of petroleum-contaminated soil mainly includes microbial remediation, plant-microbial remediation, and phytoremediation [97, 98].

BC, a black organic material produced during pyrolysis by heating at high temperature with little or no oxygen, is a valuable soil amendment. It affects the global carbon cycle, as well as improving the soil characteristics such as the microbial activity, aeration, water holding capacity, soil fertility, and reduction of leaching soluble nutrients [99]. This amendment results in helpful effects on plant growth and crop yield, being a promising amendment for remediation of oil contaminated soils. In addition, BCs have an excellent potential to absorb inorganic and organic contaminants, thus, possess the capacity for soil remediation [100–102]. During the last decade, various applications of BC have been considered and much research has been done in the field of remediation of petroleum-contaminated soil by BC with various bioremediation technologies (Table 5).

Different factors influencing hydrocarbon degradation have been reported by many authors and summarized on Table 6.

The removal mechanisms are often governed by the interactions of these pollutants with various attributes of BC [128]. For evaluating the removal efficiency of the contaminants by BCs, the identification of the underlying mechanisms of the adsorption process is needed. The adsorption behavior of BC for different contaminants (i.e., heavy metals, organic pollutants and other pollutants) are different and well correlated with the properties of contaminants [129]. In addition, the adsorption mechanism may also depend on various properties of BC including surface functional groups, specific surface area, porous structure and mineral components. The various mechanisms proposed for the interaction of BC with organic contaminants are summarized in Fig. 8.

The overall mechanisms of BC interactions with metals are given in Fig. 9.

BC has porous structure, high surface area and negative surface charge density that enables it for environmental remediation, water retention and nutrient affinity [81,131]. Recently, there is an increasing interest in the application of BC for pollutant removal from soil [132]. CEC, pore size distribution and surface area are the key factors for practical applications of BC in contaminant remediation from soil/water which these factors are affected by pyrolysis temperatures. As high pyrolysis temperatures increase microporosity, surface area, and hydrophobicity, for sorption of organic contaminants the BC obtained at high pyrolysis temperatures are effective. However, the BC formed at low temperatures is more appropriate for eliminating organic and inorganic pollutants by precipitation, electrostatic interaction and oxygen containing functional groups [130]. Galitskaya et al. [133] suggested that using of BC for remediation of oil-polluted soil may be useful in regions where remediation time is limited. Since BC using lead to alteration of soil conditions and stimulation of microorganisms which are able to

Table 5
The remediation of petroleum-contaminated soil by BC with various bioremediation technologies.

Feedstocks	Types of pollutants	Technologies	Total concentration of pollutants in soil	Degradation rate/removal rate	Refs
Corncob	TPH	Microbial	100 g/L	59.3–70.7%	[103]
Straw				48.7–58.3%	
Sawdust				45.9–57.4%	
Rice straw	PAHs	Microbial	857 ± 162 mg/kg	40.00%–58.84%	[104]
Walnut shells	PAHs	Microbial	Lower than 0.5 mg/kg		[105]
Corn cobs					
Corn stems					
Rice straw					
Wheat straw	PAHs	Microbial	3596.36 ± 80.38 mg/kg	69.95%	[106]
Rice straw	TPH	Microbial	3000 mg/kg	84.8%	[107]
Rice straw	PAHs	Microbial	677 mg/kg	62.6 ± 3.2%	[108]
Clover	TPH	Plant-microbial	16.79 mg/kg	18.6%	[109]
Mallow	TPH			45.5%	
Wheat straws	TPHs	Phytoremediation	10,000 mg/kg (1.0%, Woil/Wdry soil)	45.83%	[110]
Wheat straws	PAHs			30.34%	
Rice husk	TPHs	Phytoremediation	30 g/kg	35.1%	[111]
Corn stalks	PAHs	Phytoremediation	100 mg/kg	12–18%	[112]

Table 6
Parameters affecting bioremediation of PHC pollutants in soil.

Parameters	Effect	Refs
Oxygen availability	• Aerobic condition	
Soil pH	• pH is between 6 and 9	[113]
	• Enzymes and microorganisms show pH-dependent activity	[98]
Temperature	• Highest degradation rates that generally happen in the range 30–40°C in soil environments	[114]
	• Salinity and high temperature disrupt microbial growth and their products	[115]
	• Pollutants are more persistent at lower temperature	[98]
	• Oil viscosity increases at low temperatures, while volatility of the toxic low molecular weight hydrocarbons decrease, the onset of biodegradation is delayed	[116]
	• Temperature impacts the hydrocarbons solubility. Hydrocarbon biodegradation can occur over a wide range of temperatures, but the biodegradation rate generally decreases with decreasing temperature	[117]
	• Elevated temperature increases solubility of hydrocarbon pollutants, decreases viscosity and transfers long chain n-alkanes from solid phase to water phase	[118]
Oxidation or reduction potential	• Concentrations and ratios of electron donors/acceptors determine pathways and efficiency of degradation	[113]
Types of Pollutants	• The pollutants as substrates must be available and accessible either to microorganisms or to their extra cellular enzymes for metabolism to occur	[119]
	• Biodegradability of hydrocarbons can be ranked as: linear alkanes > branched alkanes > low-molecular-weight alkylaromatics > monoaromatics > cyclic alkanes > polyaromatics_asphaltenes	[120]
	• Some compounds, such as the high molecular weight PAHs, may not be degraded at all	[121]
Water content	• Pollutants transport and the degraded products, pollutants degradation	[98]
Nutrients	• To stimulate microbial degradation, nutrients in the form offertilizers, slow release and oleophilic are added	[122]
	• The acceleration of microbial turnover of chemical pollutants generally depends on the supply of carbon nutrients	[123]
Microbial communities	• A successful strategy for in situ bioremediation can be the combination, in a single bacterial strain or in a syntrophic bacterial consortium, of different degrading abilities with genetic traits that provide selective advantages in a given environment	[124]
Organic matter	• Effect degradation and entrapment or adsorption	[125]
Site condition	• The soil properties and the indigenous soil microbial population affect the degree of biodegradation	[126]
Co-contaminant and the presence of possible inhibitors	• In order to find out whether bioremediation itself is suitable or not	[127]

degrade hydrocarbons. Therefore, BC causes acceleration of hydrocarbon decomposition as compared with remediation by means of mixing and moistening. During the winter, PHC degradation slows considerably which substantially raises the soil remediation time in Arctic farmlands. Karppinen et al. [134] revealed that even in frozen soils BC application lead to increase PHC degradation, but further investigation is required. Kong et al. [106] used wheat straw and sawdust BCs synthesized at 300 °C and 500 °C for remediation petroleum-contaminated soil.

The BC application has proven to be a sustainable and successful and solution for the remediation of contaminated by PAHs [104,132,135].

The results of the molecular analysis showed that BC addition induced an enrichment of PAHs degraders. The way of preparing BC and the nature of PAHs are the factors impressed the bioavailability of PAHs. Zand & Grathwohl [136] studied the influence of BC particle size on leaching behavior of PAHs in BC amended soil. Based on the data, using of pulverized BC and activated carbon was favorable for immobilizing PAHs and lead to enhance sorption of PAHs compounds. So leaching PAHs from soil into water reduced. Agarry et al. [137] investigated the comparative potential effects of plantain peel-derived BC and commercial activated carbon with different particle sizes and dosage to stimulate petroleum hydrocarbon biodegradation in soil. They suggested that soil amendment by using BC possoss potential to be a low cost, effective, eco-friendly and fairly new treatment technologies to reduce soil contaminated with toxic organic compounds. Wang et al. [138] reported that petroleum-contaminated soil can be remediated by the collective use of BC and nutrients. They investigated the capacity of BC, bulrush straw powder, and BC in combination with nutrients such as phosphorus (P) and nitrogen (N) for the remediation efficiency of petroleum-contaminated soil. According to the data, the degradation of total petroleum hydrocarbons (TPHs) and the main hydrocarbon fractions was further when BC used alone and in combination with nutrients than unamended soil as the control treatment and soil amended with bulruh straw powder. Compared to the other treatments using BC in combination with nutrients lead to the intensities of the n-alkane fraction, C27–C29 steranes, moreover, the C33–C35 homohopanes were significantly reduced. In the other study, to remediation of PAHs from contaminated soil, the effect of soil amendments of lignocellulosic substrate, wheat straw BC, combination of lignocellulosic substrate and wheat straw BC, and combination of lignocellulosic substrate, wheat straw BC and surfactant Brij30 was assessed by Cao et al. [139]. The results showed that removal rates of PAHs were influenced by the related BC remediation and effects of BC on activities of dehydrogenase and polyphenol oxidase were different, moreover, lignocellulosic substrate and combination of lignocellulosic substrate, wheat straw BC and surfactant Brij30 enhance the soil microbial activity in associated BC remediation. Qin et al. [107] studied the effects of rice straw BC on soil contaminant biodegradation and suggested that not only BC could absorb petroleum metabolites and decreased the soil toxicity but also it could improve the biodegradation and displayed no appreciable negative effect on soil microorganisms. Barati et al. [140] investigated and compared the efficiency of poultry manure and its BC for the degradation of petroleum hydrocarbons in polluted soils using barley plant at different TPHs levels during five months under greenhouse conditions. Comparing the two amendments indicated that the highest microbial respiration, dry weight and TPHs degradation potential were observed in PMB treatment soil. The effects of plants growth (oat and barley) and BC (poultry manure) on the degradation of total PHC and microbial respiration rate in contaminated soil were also investigated by Barati et al. [140]. This finding suggests that the microbial respiration rate average and the percentage of TPHs reduction in BC-treated soil significantly improved by 37.73 and 21.76% for barley and 45.18 and 20.36% for oat plant in comparison with non-BC treatments, respectively. In contrast, the results of Saum et al. [112] study showed that BC addition to petroleum-contaminated soils did not improve the rate of bioremediation. While using of plants (mesquite tree seedlings), compost and combinations of these treatments improved phytoremediation of petroleum-contaminated soil. In the earlier study, hydrocarbon removal in petroleum-contaminated soil using a combination of BC amendment and phytoremediation with ryegrass was studied by Han et al. [110]. The experiment was conducted by different bioremediation strategies such as combination of ryegrass and BC, phytoremediation with ryegrass, BC amendment as well as natural attenuation. The data showed that the number of microorganisms and the removal of TPHs could be remarkably increased by planting grass in the soil. Compared with polycyclic aromatic hydrocarbons, the remediation rate of total n-alkanes was further in TPHs. The BC amendment did not lead to the meaningful promotion in removal of TPH. However, it

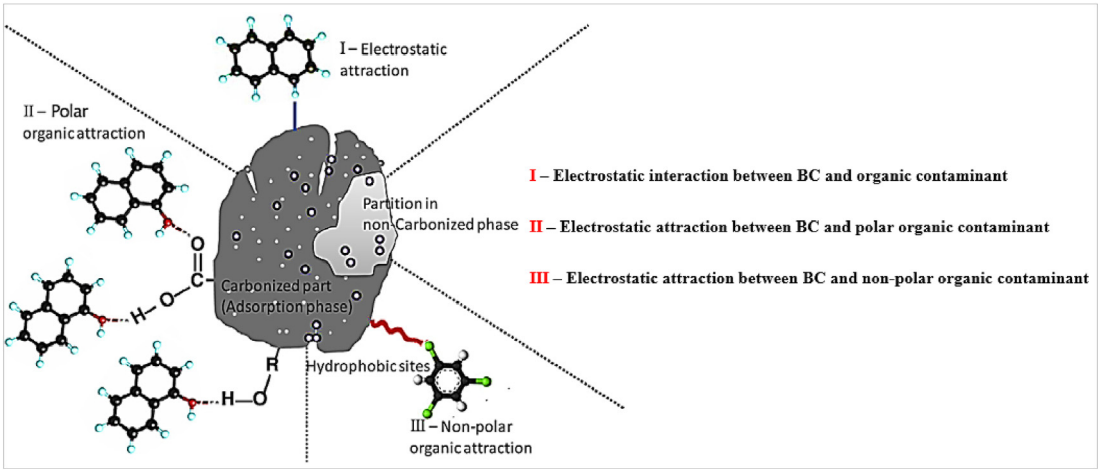


Fig. 8. Mechanisms of the interactions of BC with organic contaminants (circles on BC particle show partition or uptake) [130].

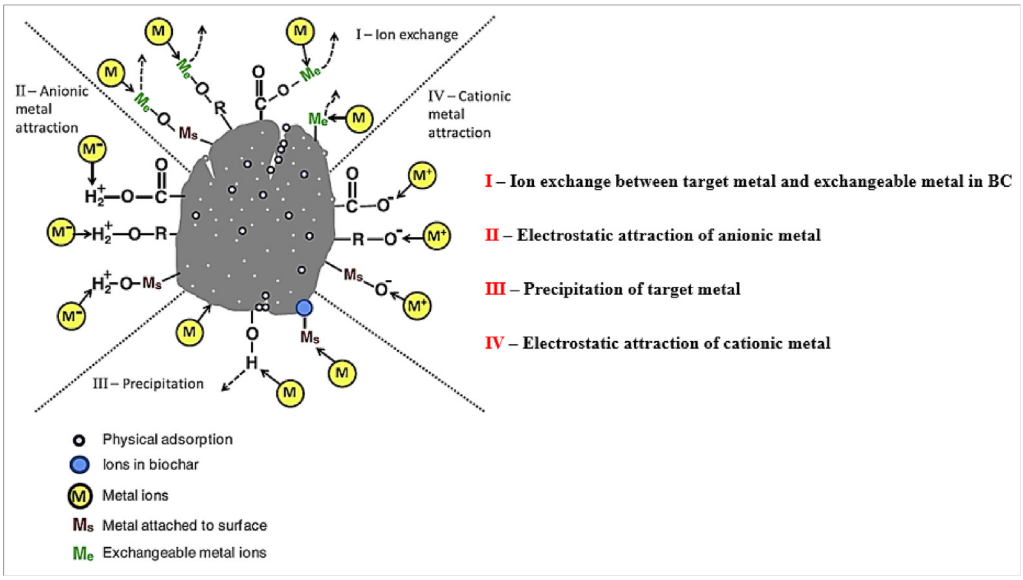


Fig. 9. Postulated mechanisms of BC interactions with inorganic contaminants (circles on BC particle show physical uptake) [130].

indicated a clear negative effect on the ryegrass growth and the TPHs removal by ryegrass. After the BC amendment, the TPHs removal rate was lower. This obviously confirms that planting ryegrass is a nominal remediation method, while the BC amendment can not be appropriate for the phytoremediation of the petroleum-contaminated soil. The efficiency of the hydrocarbon bioremediation strategies is pertaining on different paramaters, including degree and depth of pollution, pollutant bioavailability, nutrient accessibility in the target environment, pollutant nature, and so on [141]. Kołtowski et al. [142] offered that the activation of BC (using the microwave, carbon dioxide or superheated steam) caused the increase in the surface area of BC subjected (specially BC activated with superheated steam) and an increase of the effectiveness of immobilization of PAHs. Xiong et al. [108] suggested that the microbe-BC composite had a good potential for degrading phenanthrene, fluoranthene, and pyrene. They improved PAHs biodegradation in historically contaminated soil by M. gilvum inoculated BC. In the earlier study, in order to develop a new method for remediation of PHC contaminates, salt tolerant Corynebacterium variabile HRJ4 and BC were used by Zhang et al. [143]. Results indicate that the bacteria involving the BC were very efficient in degradation of PAHs (PYR, NAP) mixture and n-alkanes (C16, C18, C19, C26, C28). As results, the higher degradation of THPs up to 78.9% after 7-day incubation caused by the immobilization of C. variable HRJ4 with BC in comparison with the free leaving bacteria. Moreover, immobilization of hydrocarbon degraders results in an extra influence of BC on microbial respiration and phytotoxicity in soil. This could be due to differences in metabolic pathways of introduced and indigenous microbes.

4. Conclusion

Petroleum is one of the most important sources of energy and a raw material for many chemical products. The activities in the petroleum industry extend from the exploration stage to production, transportation as well as refining, can cause some environmental problems to groundwater, soil and air. Soil contaminated by petroleum compounds is a global environmental problem that has gained more attention in the recent years. BC has significant properties that make it proper for the soil remediation, such as porous structure, resistance to degradation, high internal surface area and negative surface charge density. Moreover, these properties of the BC improve the physical and chemical characteristics of soil, including the soil structure, cation exchange capacity, water holding capacity, and nutrients. The purpose of this review is to summarize the recent research of BC to further find out the specific properties of BC, emphasize the application of BC for soil remediation.

Feedstock properties and production conditions are two principal classification of factors determining the properties and yield of BC. These factors can lead to BC with different characteristics and therefore different impacts on soil remediation.

Declaration of competing interest

We confirm that all of the authors of this manuscript have no pecuniary or other personal interest, direct or indirect, in any matter that raises or my raise a conflict in this manuscript.

References

[1] X. Yang, K. Iris, D.C. Tsang, V.L. Budarin, J.H. Clark, K.C.-W. Wu, A.C. Yip, B. Gao, S.S. Lam, Y.S. Ok, Ball-milled, solvent-free Sn-functionalisation of wood waste biochar for sugar conversion in food waste valorisation, J. Clean. Prod. (2020) 122300.

[2] S. Salehi, S. Alijani, M. Anbia, Enhanced adsorption properties of zirconium modified chitosan-zeolite nanocomposites for vanadium ion removal, Int. J. Biol. Macromol. 164 (2020) 105–120.

[3] C. Singh, S. Tiwari, J.S. Singh, Biochar: a sustainable tool in soil pollutant bioremediation, in: Bioremediation of Industrial Waste for Environmental Safety, Springer, 2020, pp. 475–494.

[4] S. Salehi, M. Hosseinifard, Optimized removal of phosphate and nitrate from aqueous media using zirconium functionalized nanochitosan-graphene oxide composite, Cellulose 27 (2020) 8859–8883.

[5] K.-W. Jung, S.Y. Lee, J.-W. Choi, Y.J. Lee, A facile one-pot hydrothermal synthesis of hydroxyapatite/biochar nanocomposites: adsorption behavior and mechanisms for the removal of copper (II) from aqueous media, Chem. Eng. J. 369 (2019) 529–541.

[6] N. Gujre, A. Soni, L. Rangan, D.C. Tsang, S. Mitra, Sustainable improvement of soil health utilizing biochar and arbuscular mycorrhizal fungi: a review, Environ. Pollut. (2020) 115549.

[7] R. Brown, Biochar Production Technology, Biochar for environmental management: Science and technology, 2009, pp. 127–146.

[8] C. Senthil, C.W. Lee, Biomass-derived biochar materials as sustainable energy sources for electrochemical energy storage devices, Renew. Sustain. Energy Rev. (2020) 110464.

[9] Z. Kwoczynski, J. Čmelík, Characterization of biomass wastes and its possibility of agriculture utilization due to biochar production by torrefaction process, J. Clean. Prod. 280 (2021) 124302.

[10] L. El-Bassi, A.A. Azzaz, S. Jellali, H. Akrout, E.A.N. Marks, C.M. Ghimbeu, M. Jeguirim, Application of olive mill waste-based biochars in agriculture: impact on soil properties, enzymatic activities and tomato growth, Sci. Total Environ. 755 (2021) 142531.

[11] S.K. Das, G.K. Ghosh, R. Avasthe, Applications of biomass derived biochar in modern science and technology, Environ. Technol. Innov. 21 (2021) 101306.

[12] N. Kamali, A.R. Mehrabadi, M. Mirabi, M.A. Zahed, Comparison of micro and nano MgO-functionalized vinasse biochar in phosphate removal: micro-nano particle development, RSM optimization, and potential fertilizer, J. Water Process Eng. (2020) 101741.

[13] A. Sajjad, F. Jabeen, M. Farid, Q. Fatima, A. Akbar, Q. Ali, I. Hussain, U. Iftikhar, S. Farid, H.K. Ishaq, Biochar: a sustainable product for remediation of contaminated soils, in: Plant Ecophysiology and Adaptation under Climate Change: Mechanisms and Perspectives II, Springer, 2020, pp. 787–799.

[14] S. Aziz, M.I. Ali, U. Farooq, A. Jamal, F.-J. Liu, H. He, H. Guo, M. Urynowicz, Z. Huang, Enhanced bioremediation of diesel range hydrocarbons in soil using biochar made from organic wastes, Environ. Monit. Assess. 192 (2020) 1–14.

[15] L. Wang, L. Chen, D.C. Tsang, H.W. Kua, J. Yang, Y.S. Ok, S. Ding, D. Hou, C.S. Poon, The roles of biochar as green admixture for sediment-based construction products, Cement Concr. Compos. 104 (2019) 103348.

[16] M. Kumar, X. Xiong, Z. Wan, Y. Sun, D.C. Tsang, J. Gupta, B. Gao, X. Cao, J. Tang, Y.S. Ok, Ball milling as a mechanochemical technology for fabrication of novel biochar nanomaterials, Bioresour. Technol. (2020) 123613.

[17] N. Kamali, A.R. Mehrabadi, M. Mirabi, M.A. Zahed, Synthesis of vinasse-dolomite nanocomposite biochar via a novel developed functionalization method to recover phosphate as a potential fertilizer substitute, Front. Environ. Sci. Eng. 14 (2020).

[18] B.A. Mohamed, N. Ellis, C.S. Kim, X. Bi, A.E.-r. Emam, Engineered biochar from microwave-assisted catalytic pyrolysis of switchgrass for increasing water-holding capacity and fertility of sandy soil, Sci. Total Environ. 566 (2016) 387–397.

[19] N.S. Mirhosseinian, M. Anbia, S. Salehi, Preparation and characterization of superhydrophobic melamine and melamine-derived carbon sponges modified with reduced graphene oxide–TiO 2 nanocomposite as oil absorbent materials, J. Mater. Sci. 55 (2020) 1536–1552.

[20] M. Tajer, M. Anbia, S. Salehi, Fabrication of polyacrylonitrile hybrid nanofiber scaffold containing activated carbon by electrospinning process as nanofilter media for SO_2, CO_2 and CH_4 adsorption, Environ. Prog. Sustain. 40 (2021).

[21] M. Anbia, N. Kakoli Khataei, S. Salehi, Ordered nanoporous carbon (CMK-3) coated fiber for solid-phase microextraction of benzene and chlorobenzenes in water samples, Advances in Environmental Technology 4 (2018) 13–22.

[22] J. Lee, K.-H. Kim, E.E. Kwon, Biochar as a catalyst, Renew. Sustain. Energy Rev. 77 (2017) 70–79.

[23] F. Eshraghi, M. Anbia, S. Salehi, Dative post synthetic methods on SBUs of MWCNT@MOFs hybrid composite and its effect on CO_2 uptake properties, J. Environ. Chem. Eng. 5 (2017) 4516–4523.

[24] S. Zohdi, M. Anbia, S. Salehi, Improved CO_2 adsorption capacity and CO_2/CH_4 and CO_2/N_2 selectivity in novel hollow silica particles by modification with multi-walled carbon nanotubes containing amine groups, Polyhedron 166 (2019) 175–185.

[25] M. Anbia, S. Salehi, Adsorption of methyl-orange from aqueous solution onto nanoporous silica materials, Sci. Iran. 21 (2014) 2036–2048.

[26] M. Zeeshan, W. Ahmad, F. Hussain, W. Ahamd, M. Numan, M. Shah, I. Ahmad, Phytostabalization of the heavy metals in the soil with biochar applications, the impact on chlorophyll, carotene, soil fertility and tomato crop yield, J. Clean. Prod. 255 (2020) 120318.

[27] L. Giagnoni, A. Maienza, S. Baronti, F.P. Vaccari, L. Genesio, C. Taiti, T. Martellini, R. Scodellini, A. Cincinelli, C. Costa, S. Mancuso, G. Renella, Long-term soil biological fertility, volatile organic compounds and chemical properties in a vineyard soil after biochar amendment, Geoderma 344 (2019) 127–136.

[28] J. Zhang, J. Jin, M. Wang, R. Naidu, Y. Liu, Y.B. Man, X. Liang, M.H. Wong, P. Christie, Y. Zhang, C. Song, S. Shan, Co-pyrolysis of sewage sludge and rice husk/bamboo sawdust for biochar with high aromaticity and low metal mobility, Environ. Res. 191 (2020) 110034.

[29] M.J. Ahmed, B.H. Hameed, Insight into the co-pyrolysis of different blended feedstocks to biochar for the adsorption of organic and inorganic pollutants: a review, J. Clean. Prod. 265 (2020) 121762.

[30] M.I. Inyang, B. Gao, Y. Yao, Y. Xue, A. Zimmerman, A. Mosa, P. Pullammanappallil, Y.S. Ok, X. Cao, A review of biochar as a low-cost adsorbent for aqueous heavy metal removal, Crit. Rev. Environ. Sci. Technol. 46 (2016) 406–433.

[31] M. Saber, B. Nakhshiniev, K. Yoshikawa, A review of production and upgrading of algal bio-oil, Renew. Sustain. Energy Rev. 58 (2016) 918–930.

[32] S. Salehi, M. Anbia, Highly efficient CO_2 capture with a metal–organic framework-derived porous carbon impregnated with polyethyleneimine, Appl. Organomet. Chem. (2018), e4390.

[33] S. Salehi, M. Anbia, F. Razavi, Improving CO_2/CH_4 and CO_2/N_2 adsorptive selectivity of Cu-BTC and MOF-derived nanoporous carbon by modification with nitrogen-containing groups, Environ. Prog. Sustain. 39 (2020) 13302.

[34] S. Salehi, S. Mandegarzad, M. Anbia, Preparation and characterization of metal organic framework-derived nanoporous carbons for highly efficient removal of vanadium from aqueous solution, J. Alloys Compd. 812 (2020) 152051.

[35] M.I. Jahirul, M.G. Rasul, A.A. Chowdhury, N. Ashwath, Biofuels production through biomass pyrolysis—a technological review, Energies 5 (2012) 4952–5001.

[36] A.K. Patel, Land applications of biochar: an emerging area, in: Waste to Wealth, Springer, 2018, pp. 171–197.

[37] A.D. Igalavithana, S. Mandal, N.K. Niazi, M. Vithanage, S.J. Parikh, F.N. Mukome, M. Rizwan, P. Oleszczuk, M. Al-Wabel, N. Bolan, Advances and future directions of biochar characterization methods and applications, Crit. Rev. Environ. Sci. Technol. 47 (2017) 2275–2330.

[38] T. Sizmur, T. Fresno, G. Akgül, H. Frost, E. Moreno-Jiménez, Biochar modification to enhance sorption of inorganics from water, Bioresour. Technol. 246 (2017) 34–47.

[39] L. Zhao, X. Cao, O. Mašek, A. Zimmerman, Heterogeneity of biochar properties as a function of feedstock sources and production temperatures, J. Hazard Mater. 256 (2013) 1–9.

[40] F. Liao, L. Yang, Q. Li, A. Muhammad, J. Xue, D. Huang, Y. Li, Factor analysis and comprehensive evaluation for quality of biochar derived from different biomass, J. S. China Agric. Univ. 40 (2019) 29–37.

[41] T.M.E. Shareef, B. Zhao, The fundamentals of biochar as a soil amendment tool and management in agriculture scope: an overview for farmers and gardeners, J. Agric. Chem. Environ. 6 (2016) 38–61.

[42] X. Yang, H. Wang, P.J. Strong, S. Xu, S. Liu, K. Lu, K. Sheng, J. Guo, L. Che, L. He, Thermal properties of biochars derived from waste biomass generated by agricultural and forestry sectors, Energies 10 (2017) 469.

[43] M.K. Rafiq, R.T. Bachmann, M.T. Rafiq, Z. Shang, S. Joseph, R. Long, Influence of pyrolysis temperature on physico-chemical properties of corn stover (Zea mays L.) biochar and feasibility for carbon capture and energy balance, PloS One 11 (2016), e0156894.

[44] I.F. Titiladunayo, A.G. McDonald, O.P. Fapetu, Effect of temperature on biochar product yield from selected lignocellulosic biomass in a pyrolysis process, Waste Biomass Valorization 3 (2012) 311–318.

[45] C. Zhao, A. Chen, E. Jiang, L. Qin, Pyrolysis of industrial waste lignin: analysis of product yields and character, Energy Sources Part A 39 (2017) 458–464.

[46] H. Li, S.A.A. Mahyoub, W. Liao, S. Xia, H. Zhao, M. Guo, P. Ma, Effect of pyrolysis temperature on characteristics and aromatic contaminants adsorption behavior of magnetic biochar derived from pyrolysis oil distillation residue, Bioresour. Technol. 223 (2017) 20–26.

[47] Q. Wang, Z. Lai, J. Mu, D. Chu, X. Zang, Converting industrial waste cork to biochar as Cu (II) adsorbent via slow pyrolysis, Waste Manag. 105 (2020) 102–109.

[48] S.M. Yakout, Physicochemical characteristics of biochar produced from rice straw at different pyrolysis temperature for soil amendment and removal of organics, Proc. Natl. Acad. Sci. India: Physical Sciences 87 (2017) 207–214.

[49] B. Zhao, D. O'Connor, J. Zhang, T. Peng, Z. Shen, D.C. Tsang, D. Hou, Effect of pyrolysis temperature, heating rate, and residence time on rapeseed stem derived biochar, J. Clean. Prod. 174 (2018) 977–987.

[50] J.K. Sarkar, Q. Wang, Different pyrolysis process conditions of South asian waste coconut shell and characterization of gas, Bio-Char, and Bio-Oil, Energies 13 (2020) 1970.
[51] O. Mašek, P. Brownsort, A. Cross, S. Sohi, Influence of production conditions on the yield and environmental stability of biochar, Fuel 103 (2013) 151–155.
[52] A. Hmid, D. Mondelli, S. Fiore, F.P. Fanizzi, Z. Al Chami, S. Dumontet, Production and characterization of biochar from three-phase olive mill waste through slow pyrolysis, Biomass Bioenergy 71 (2014) 330–339.
[53] T. Mimmo, P. Panzacchi, M. Baratieri, C. Davies, G. Tonon, Effect of pyrolysis temperature on miscanthus (Miscanthus× giganteus) biochar physical, chemical and functional properties, Biomass Bioenergy 62 (2014) 149–157.
[54] D. Angın, Effect of pyrolysis temperature and heating rate on biochar obtained from pyrolysis of safflower seed press cake, Bioresour. Technol. 128 (2013) 593–597.
[55] J.J. Manyà, M. Azuara, J.A. Manso, Biochar production through slow pyrolysis of different biomass materials: seeking the best operating conditions, Biomass Bioenergy 117 (2018) 115–123.
[56] A. Demirbas, G. Arin, An overview of biomass pyrolysis, Biomass Bioenergy 24 (2002) 471–482.
[57] C. Liu, X. Liu, Y. He, X. An, D. Fan, Z. Wu, Microwave-assisted catalytic pyrolysis of apple wood to produce biochar: Co-pyrolysis behavior, pyrolysis kinetics analysis and evaluation of microbial carriers, Bioresour. Technol. 320 (2021) 124345.
[58] M. Guiotoku, C.d.F. Maia, C.R. Rambo, D. Hotza, Synthesis of Carbon-Based Materials by Microwave Hydrothermal Processing, Embrapa Florestas-Capítulo em livro científico (ALICE), 2011.
[59] Y.-F. Huang, P.-T. Chiueh, W.-H. Kuan, S.-L. Lo, Microwave pyrolysis of lignocellulosic biomass: heating performance and reaction kinetics, Energy 100 (2016) 137–144.
[60] F. Motasemi, M.T. Afzal, A review on the microwave-assisted pyrolysis technique, Renew. Sustain. Energy Rev. 28 (2013) 317–330.
[61] B.-J. Lin, W.-H. Chen, Sugarcane bagasse pyrolysis in a carbon dioxide atmosphere with conventional and microwave-assisted heating, Front. Energy Res. 3 (2015) 4.
[62] H. Kadlimatti, B. Raj Mohan, M. Saidutta, Microwave-assisted pyrolysis of food waste: optimization of fixed carbon content using response surface methodology, Biofuels (2019) 1–8.
[63] X. Zhang, K. Rajagopalan, H. Lei, R. Ruan, B.K. Sharma, An overview of a novel concept in biomass pyrolysis: microwave irradiation, Sustain. Energy Fuels 1 (2017) 1664–1699.
[64] P. Wen, Z. Wu, Y. Han, G. Cravotto, J. Wang, B.-C. Ye, Microwave-assisted synthesis of a novel biochar-based slow-release nitrogen fertilizer with enhanced water-retention capacity, ACS Sustain. Chem. Eng. 5 (2017) 7374–7382.
[65] T. Haeldermans, L. Campion, T. Kuppens, K. Vanreppelen, A. Cuypers, S. Schreurs, A comparative techno-economic assessment of biochar production from different residue streams using conventional and microwave pyrolysis, Bioresour. Technol. 318 (2020) 124083.
[66] J. Wang, S. Wang, Preparation, modification and environmental application of biochar: a review, J. Clean. Prod. 227 (2019) 1002–1022.
[67] L. Bowlby, G. Saha, M. Afzal, Synthesis of high-surface-area biochar particles using microwave pyrolysis technique, Can. Soc. Bioeng. Annu. Gen. Meet. (2017) 1–10.
[68] M. Miura, H. Kaga, A. Sakurai, T. Kakuchi, K. Takahashi, Rapid pyrolysis of wood block by microwave heating, J. Anal. Appl. Pyrolysis 71 (2004) 187–199.
[69] V.L. Budarin, J.H. Clark, B.A. Lanigan, P. Shuttleworth, D.J. Macquarrie, Microwave assisted decomposition of cellulose: a new thermochemical route for biomass exploitation, Bioresour, Technol. 101 (2010) 3776–3779.
[70] S. Ethaib, R. Omar, S.M.M. Kamal, D.R. Awang Biak, S.L. Zubaidi, Microwave-Assisted pyrolysis of biomass waste: a mini review, Processes 8 (2020) 1190.
[71] C. Nzediegwu, M. Arshad, A. Ulah, M.A. Naeth, S.X. Chang, Fuel, thermal and surface properties of microwave-pyrolyzed biochars depend on feedstock type and pyrolysis temperature, Bioresour. Technol. 320 (2021) 124282.
[72] T. Xie, K.R. Reddy, C. Wang, E. Yargicoglu, K. Spokas, Characteristics and applications of biochar for environmental remediation: a review, Crit. Rev. Environ. Sci. Technol. l45 (2015) 939–969.
[73] J.K. Saini, R. Saini, L. Tewari, Lignocellulosic agriculture wastes as biomass feedstocks for second-generation bioethanol production: concepts and recent developments, 3 Biotech 5 (2015) 337–353.
[74] C.M. Welker, V.K. Balasubramanian, C. Petti, K.M. Rai, S. DeBolt, V. Mendu, Engineering plant biomass lignin content and composition for biofuels and bioproducts, Energies 8 (2015) 7654–7676.
[75] F. Abnisa, A. Arami-Niya, W.W. Daud, J. Sahu, Characterization of bio-oil and bio-char from pyrolysis of palm oil wastes, BioEnergy Research 6 (2013) 830–840.
[76] W. Kwapinski, C.M. Byrne, E. Kryachko, P. Wolfram, C. Adley, J.J. Leahy, E.H. Novotny, M.H. Hayes, Biochar from biomass and waste, Waste Biomass Valorization 1 (2010) 177–189.
[77] A. Friedl, E. Padouvas, H. Rotter, K. Varmuza, Prediction of heating values of biomass fuel from elemental composition, Anal. Chim. Acta 544 (2005) 191–198.
[78] D. Wu, L. Xiao, Y. Ba, H. Wang, A. Zhang, X. Wu, M. Niu, K. Fang, The recovery of energy, nitrogen and phosphorous from three agricultural wastes by pyrolysis, Energy Procedia 105 (2017) 1263–1269.
[79] J.M. Mesa-Pérez, J.D. Rocha, L.A. Barbosa-Cortez, M. Penedo-Medina, C.A. Luengo, E. Cascarosa, Fast oxidative pyrolysis of sugar cane straw in a fluidized bed reactor, Appl. Therm. Eng. 56 (2013) 167–175.
[80] K.L. Yu, P.L. Show, H.C. Ong, T.C. Ling, J.C.-W. Lan, W.-H. Chen, J.-S. Chang, Microalgae from wastewater treatment to biochar–Feedstock preparation and conversion technologies, Energy Convers. Manag. 150 (2017) 1–13.
[81] A.U. Rajapaksha, S.S. Chen, D.C. Tsang, M. Zhang, M. Vithanage, S. Mandal, B. Gao, N.S. Bolan, Y.S. Ok, Engineered/designer biochar for contaminant removal/immobilization from soil and water: potential and implication of biochar modification, Chemosphere 148 (2016) 276–291.
[82] T. Sizmur, R. Quilliam, A.P. Puga, E. Moreno-Jiménez, L. Beesley, J.L. Gomez-Eyles, Application of biochar for soil remediation, Agricultural and environmental applications of biochar: Advances and barriers 63 (2016) 295–324.
[83] Z.-H. Yang, F. Verpoort, C.-D. Dong, C.-W. Chen, S. Chen, C.-M. Kao, Remediation of petroleum-hydrocarbon contaminated groundwater using optimized in situ chemical oxidation system: batch and column studies, Process Saf. Environ. Protect. 138 (2020) 18–26.
[84] B. Wartell, M. Boufadel, L. Rodriguez-Freire, An effort to understand and improve the anaerobic biodegradation of petroleum hydrocarbons: a literature review, Int. Biodeterior. Biodegrad. 157 (2021) 105156.
[85] S.C. Wilson, K.C. Jones, Bioremediation of soil contaminated with polynuclear aromatic hydrocarbons (PAHs): a review, Environ. Pollut. 81 (1993) 229–249.
[86] M. Blumer, Polycyclic aromatic compounds in nature, Sci. Am. 234 (1976) 34–45.
[87] S. Kuppusamy, N.R. Maddela, M. Megharaj, K. Venkateswarlu, An overview of total petroleum hydrocarbons, in: Total Petroleum Hydrocarbons, Springer, 2020, pp. 1–27.
[88] K. Martinez-Swatson, E. Mihály, C. Lange, M. Ernst, M. Dela Cruz, M.J. Price, T.N. Mikkelsen, J.H. Christensen, N. Lundholm, N. Rønsted, Biomonitoring of polycyclic aromatic hydrocarbon deposition in Greenland using historical moss herbarium specimens shows a decrease in pollution during the 20th century, Front. Plant Sci. 11 (2020) 1085.
[89] A.R. Autry, G.M. Ellis, Bioremediation: an effective remedial alternative for petroleum hydrocarbon-contaminated soil, Environ. Prog. 11 (1992) 318–323.
[90] K. Urum, S. Grigson, T. Pekdemir, S. McMenamy, A comparison of the efficiency of different surfactants for removal of crude oil from contaminated soils, Chemosphere 62 (2006) 1403–1410.
[91] J. Pinedo, R. Ibáñez, J.P.A. Lijzen, Á. Irabien, Assessment of soil pollution based on total petroleum hydrocarbons and individual oil substances, J. Environ. Manag. 130 (2013) 72–79.
[92] S.A. Hoang, D. Lamb, B. Seshadri, B. Sarkar, G. Choppala, M.B. Kirkham, N.S. Bolan, Rhizoremediation as a green technology for the remediation of petroleum hydrocarbon-contaminated soils, J. Hazard Mater. 401 (2021) 123282.
[93] S. Wang, Y. Xu, Z. Lin, J. Zhang, N. Norbu, W. Liu, The harm of petroleum-polluted soil and its remediation research, in: AIP Conference Proceedings, AIP Publishing LLC, 2017, 020222.
[94] W. Chen, F. Wang, L. Zeng, Q. Li, Bioremediation of petroleum-contaminated soil by semi-aerobic aged refuse biofilter: optimization and mechanism, J. Clean. Prod. (2020) 125354.
[95] K. Zhang, S. Wang, P. Guo, S. Guo, Characteristics of organic carbon metabolism and bioremediation of petroleum-contaminated soil by a mesophilic aerobic biopile system, Chemosphere 264 (2021) 128521.
[96] S. Guo, R. Fan, T. Li, N. Hartog, F. Li, X. Yang, Synergistic effects of bioremediation and electrokinetics in the remediation of petroleum-contaminated soil, Chemosphere 109 (2014) 226–233.
[97] E. Riser-Roberts, Remediation of Petroleum Contaminated Soils: Biological, Physical, and Chemical Processes, CRC press, 2020.
[98] M.D. Yuniati, Bioremediation of petroleum-contaminated soil: a Review, in: IOP Conference Series: Earth and Environmental Science, 2018, 1755-1315.
[99] P. Henner, M. Schiavon, J.-L. Morel, E. Lichtfouse, Polycyclic Aromatic Hydrocarbon (PAH) Occurrence and Remediation Methods, 1997.
[100] S. Gupta, H.W. Kua, Factors determining the potential of biochar as a carbon capturing and sequestering construction material: critical review, J. Mater. Civ. Eng. 29 (2017), 04017086.
[101] S. Nanda, S.N. Reddy, S.K. Mitra, J.A. Kozinski, The progressive routes for carbon capture and sequestration, Energy Sci. Eng. 4 (2016) 99–122.
[102] R.B. Fidel, D.A. Laird, M.L. Thompson, M. Lawrinenko, Characterization and quantification of biochar alkalinity, Chemosphere 167 (2017) 367–373.
[103] H.-Y. Ren, Z.-J. Wei, Y. Wang, Y.-P. Deng, M.-Y. Li, B. Wang, Effects of biochar properties on the bioremediation of the petroleum-contaminated soil from a shale-gas field, Environ. Sci. Pollut. Res. 27 (2020) 36427–36438.
[104] G. Zhang, L. He, X. Guo, Z. Han, L. Ji, Q. He, L. Han, K. Sun, Mechanism of biochar as a biostimulation strategy to remove polycyclic aromatic hydrocarbons from heavily contaminated soil in a coking plant, Geoderma 375 (2020) 114497.
[105] G. Zhang, X. Guo, Y. Zhu, X. Liu, Z. Han, K. Sun, L. Ji, Q. He, L. Han, The effects of different biochars on microbial quantity, microbial community shift, enzyme activity, and biodegradation of polycyclic aromatic hydrocarbons in soil, Geoderma 328 (2018) 100–108.
[106] L. Kong, Y. Gao, Q. Zhou, X. Zhao, Z. Sun, Biochar accelerates PAHs biodegradation in petroleum-polluted soil by biostimulation strategy, J. Hazard Mater. 343 (2018) 276–284.
[107] G. Qin, D. Gong, M.-Y. Fan, Bioremediation of petroleum-contaminated soil by biostimulation amended with biochar, Int. Biodeterior. Biodegrad. 85 (2013) 150–155.
[108] B. Xiong, Y. Zhang, Y. Hou, H.P.H. Arp, B.J. Reid, C. Cai, Enhanced biodegradation of PAHs in historically contaminated soil by M. gilvum inoculated biochar, Chemosphere 182 (2017) 316–324.
[109] A. Abbaspour, F. Zohrabi, V. Dorostkar, A. Faz, J.A. Acosta, Remediation of an oil-contaminated soil by two native plants treated with biochar and mycorrhizae, J. Environ. Manag. 254 (2020) 109755.
[110] T. Han, Z. Zhao, M. Bartlam, Y. Wang, Combination of biochar amendment and phytoremediation for hydrocarbon removal in petroleum-contaminated soil, Environ. Sci. Pollut. Res. 23 (2016) 21219–21228.

[111] M. Zhen, H. Chen, Q. Liu, B. Song, Y. Wang, J. Tang, Combination of rhamnolipid and biochar in assisting phytoremediation of petroleum hydrocarbon contaminated soil using Spartina anglica, Int. J. Environ. Sci. 85 (2019) 107–118.
[112] L. Saum, M.B. Jiménez, D. Crowley, Influence of biochar and compost on phytoremediation of oil-contaminated soil, Int. J. Phytoremediation 20 (2018) 54–60.
[113] N. Das, P. Chandran, Microbial degradation of petroleum hydrocarbon contaminants: an overview, Biotechnol. Res. Int. 2011 (2011).
[114] R. Bartha, I. Bossert, The Treatment and Disposal of Petroleum Wastes in Petroleum Microbiology, Macmillan, New York, NY, USA, 1984.
[115] M. Thamer, A.R. Al-Kubaisi, Z. Zahraw, H.A. Abdullah, I. Hindy, A. Abd Khadium, Biodegradation of Kirkuk Light Crude Oil by Bacillus Thuringiensis, Northern of Iraq, 2013.
[116] R.M. Atlas, Effects of temperature and crude oil composition on petroleum biodegradation, Appl. Microbiol. 30 (1975) 396–403.
[117] J.M. Foght, D.W. Westlake, W.M. Johnson, H.F. Ridgway, Environmental gasoline-utilizing isolates and clinical isolates of Pseudomonas aeruginosa are taxonomically indistinguishable by chemotaxonomic and molecular techniques, Microbios 142 (1996) 2333–2340.
[118] A.I. Okoh, Biodegradation alternative in the cleanup of petroleum hydrocarbon pollutants, Biotechnol. Mol. Biol. 1 (2006) 38–50.
[119] K.T. Semple, K.J. Doick, K.C. Jones, P. Burauel, A. Craven, H. Harms, Defining bioavailability and bioaccessibility of contaminated soil and sediment is complicated, Environ. Sci. Technol. 38 (2004) 228a–231a.
[120] J.G. Leahy, R.R. Colwell, Microbial degradation of hydrocarbons in the environment, Microbiol. Mol. Biol. Rev. 54 (1990) 305–315.
[121] R. Atlas, J. Bragg, Bioremediation of marine oil spills: when and when not–the Exxon Valdez experience, Microb. Biotechnol. 2 (2009) 213–221.
[122] M. Nikolopoulou, N. Kalogerakis, Enhanced bioremediation of crude oil utilizing lipophilic fertilizers combined with biosurfactants and molasses, Mar. Pollut. Bull. 56 (2008) 1855–1861.
[123] S.-J. Kim, D.H. Choi, D.S. Sim, Y.-S. Oh, Evaluation of bioremediation effectiveness on crude oil-contaminated sand, Chemosphere 59 (2005) 845–852.
[124] E. Díaz, Bacterial Degradation of Aromatic Pollutants: a Paradigm of Metabolic Versatility, 2004.
[125] J. Michels, T. Track, U. Gehrke, D. Sell, Biologische Verfahren zur Bodensanierung, Grun-weise Reihe des BMBF, Bonn, 2000.
[126] F.M. Bento, F.A. Camargo, B.C. Okeke, W.T. Frankenberger, Comparative bioremediation of soils contaminated with diesel oil by natural attenuation, biostimulation and bioaugmentation, Bioresour. Technol. 96 (2005) 1049–1055.
[127] J. Sabaté, M. Vinas, A. Solanas, Laboratory-scale bioremediation experiments on hydrocarbon-contaminated soils, Int. Biodeterior. Biodegrad. 54 (2004) 19–25.
[128] X. Tan, Y. Liu, G. Zeng, X. Wang, X. Hu, Y. Gu, Z. Yang, Application of biochar for the removal of pollutants from aqueous solutions, Chemosphere 125 (2015) 70–85.
[129] M. Guo, W. Song, J. Tian, Biochar-facilitated soil remediation: mechanisms and efficacy variations, Front. environ. sci. 8 (2020) 183.
[130] M. Ahmad, A.U. Rajapaksha, J.E. Lim, M. Zhang, N. Bolan, D. Mohan, M. Vithanage, S.S. Lee, Y.S. Ok, Biochar as a sorbent for contaminant management in soil and water: a review, Chemosphere 99 (2014) 19–33.
[131] K. Metcalf, Biochar for carbon sequestration: investigation and outreach, Earth Sci. 32 (2016) 33–36.
[132] F. Bianco, M. Race, S. Papirio, P. Oleszczuk, G. Esposito, The addition of biochar as a sustainable strategy for the remediation of PAH–contaminated sediments, Chemosphere (2020) 128274.
[133] P. Galitskaya, L. Akhmetzyanova, S. Selivanovskaya, Biochar-carrying hydrocarbon decomposers promote degradation during the early stage of bioremediation, Biogeosciences 13 (2016) 5739.
[134] E.M. Karppinen, K.J. Stewart, R.E. Farrell, S.D. Siciliano, Petroleum hydrocarbon remediation in frozen soil using a meat and bonemeal biochar plus fertilizer, Chemosphere 173 (2017) 330–339.
[135] L. Mohajeri, H.A. Aziz, M.H. Isa, M.A. Zahed, S. Mohajeri, Ex-situ bioremediation of crude oil in soil, a comparative kinetic analysis, Bull. Environ. Contam. Toxicol. 85 (2010) 54–58.
[136] A.D. Zand, P. Grathwohl, Enhanced immobilization of polycyclic aromatic hydrocarbons in contaminated soil using forest wood-derived biochar and activated carbon under saturated conditions, and the importance of biochar particle size, Pol. J. Environ. Stud. 25 (2016).
[137] S.E. Agarry, K.M. Oghenejoboh, B.O. Solomon, Kinetic modelling and half life study of adsorptive bioremediation of soil artificially contaminated with bonny light crude oil, J. Ecol. Eng. 16 (2015) 1–13.
[138] Y. Wang, F. Li, X. Rong, H. Song, J. Chen, Remediation of petroleum-contaminated soil using bulrush straw powder, biochar and nutrients, Bull. Environ. Contam. Toxicol. 98 (2017) 690–697.
[139] Y. Cao, B. Yang, Z. Song, H. Wang, F. He, X. Han, Wheat straw biochar amendments on the removal of polycyclic aromatic hydrocarbons (PAHs) in contaminated soil, Ecotoxicol. Environ. Saf. 130 (2016) 248–255.
[140] M. Barati, F. Bakhtiari, D. Mowla, S. Safarzadeh, Comparison of the effects of poultry manure and its biochar on barley growth in petroleum-contaminated soils, Int. J. Phytoremediation 20 (2018) 98–103.
[141] L. Mohajeri, M.A. Zahed, H. Abdul Aziz, M. Hasnain Isa, Assessment of bioaugmentation and biostimulation efficiencies for petroleum contaminated sediments, Environ. Energ. Eco. Res. 1 (2017) 89–98.
[142] M. Kołtowski, I. Hilber, T.D. Bucheli, B. Charmas, J. Skubiszewska-Zięba, P. Oleszczuk, Activated biochars reduce the exposure of polycyclic aromatic hydrocarbons in industrially contaminated soils, Chem. Eng. J. 310 (2017) 33–40.
[143] H. Zhang, J. Tang, L. Wang, J. Liu, R.G. Gurav, K. Sun, A novel bioremediation strategy for petroleum hydrocarbon pollutants using salt tolerant Corynebacterium variabile HRJ4 and biochar, Int. J. Environ. Sci. 47 (2016) 7–13.